MILITARY SPECIFICATION

MESSAGE PROCESSING AND DISTRIBUTION SYSTEM

(FOR CVA(N)-68)

1. SCOPE

1.1 This specification describes the operational and technical requirements for an electronic automatic Message Processing and Distribution System (MPDS) for the CVA(N)-68. It will be a digital store and forward message switch designed to process, store, and internally distribute incoming and outgoing digital data and narrative communications traffic. In addition, capability will be provided for generation and formatting of outgoing traffic. Messages will consist of a heading, text, and end of message in standard formats. The design of the MPDS will provide for reliable and continuous operation. It is intended that the requirements of the specification be met without exploratory development.

1.2 The system will include up to twenty small communications terminals (user terminals) in principal ship's operating stations and offices as specified by the ship's characteristics, all connected to a central message processing and storage facility in the communications center. Capability will be provided for generating, receiving, and retrieving messages. For optimum speed and effectiveness, this system shall include input/output equipment to place it "on-line" with communication receiving and transmitting channels handling traffic of record.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Test, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

FEDERAL

FED-STD-222 - Radiation Standard for Communications and Other Information Processing Equipment.

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-188 - Military Communication System Technical Standards.
MIL-STD-189 - Racks, Electrical Equipment, 19-Inch, and Associated Panels.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

DRAWINGS

MILITARY

800-2561327 - Attack Aircraft Carrier CVA(N)-68 Communication Center Arrangement of Equipment.

PUBLICATIONS

MILITARY

NAVSHIPS 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

FSC 5895

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MILITARY(cont'd.)
NAVSHIPS 0900-004-4000 - Design Guide for Integrated Circuits.
NAVSHIPS 0967-051-6040 - Input/Output Specification for the CP-642B/USQ-20(V) Computer and Related Equipment.

OPNAV
DNC-5 - U.S. Naval Communications Instructions.
ACP-117 - Allied Routing Indicator Book.
ACP-127 - Communications Instructions, Tape Relay Procedures.
JANAP-128 - Automatic Digital Network (AUTODIN) Operating Procedures.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33d Street, New York 10016).

3. REQUIREMENTS

3.1 General requirements. - The message processing and distribution system shall be in accordance with the requirements of MIL-E-16400, except as otherwise specified herein.

3.1.1 Temperature. - The equipment shall be class 4 in accordance with MIL-E-16400.

3.1.2 Material and parts. - Material and parts shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be paid to the unacceptable material requirements of MIL-E-16400.

3.1.3 Shock. - The equipment shall be designed to withstand the shock test for grade A, class I, type A equipment in the appropriate weight categories in accordance with MIL-S-901.

3.1.4 Enclosure. - The equipment shall be dripproof in accordance with MIL-STD-108.

3.1.5 Low level signaling. - The requirements of MIL-STD-188 for low level signaling shall apply.

3.1.6 Design. - The guidance of NAVSHIPS 0900-004-4000 shall be utilized in the design of the equipment in addition to the requirements of MIL-E-16400.

3.1.7 Reliability. - The required reliability level shall be determined as mean-time between failure (MTBF) calculated in accordance with procedure specified in MIL-R-22732 as modified hereinafter.

3.1.7.1 The reliability of the equipment shall be determined on a part, sub-equipment, and equipment basis. The applicable paragraphs of MIL-R-22732 and method D of NAVSHIPS 93820 shall govern the philosophies and procedures. The MTBF shall be updated quarterly with the aid of in-plant and field-failure information or both. When the last deliverable equipment has been shipped, the current MTBF shall be considered representative of all equipments. The first MTBF prediction shall be made no later than 5 months after receipt of order.

3.1.7.2 The MTBF of the equipment shall be verified by information derived from in-plant and field-failure reports received by the contractor no later than the warranty expiration date of the last deliverable item.

3.1.7.3 In the event that in-plant and field-failure information indicates that the MTBF of the equipment is less than the final prediction, the field-failure information shall be analyzed and corrective action recommended.

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3.1.7.4 In lieu of specific report requirements, the following reporting procedures shall be observed:
(a) Reliability assurance program activities shall be reported monthly by means of an informal letter report.
(b) Quarterly MTBF predictions shall be included in a monthly status report.
(c) The final MTBF prediction shall be made in a formal report which shall include a summary of program activities, a final MTBF prediction, a summary of failure reports and corrective actions, tabulations of failure rates of subequipment items, and tabulations of failure rates and stress level of parts.

3.1.8 Primary power. - The equipment shall operate within the requirements specified in MIL-E-16400, when connected to 60 or 400 cycles per second three phase, three-wire ungrounded power source. The MPDS programming shall provide for rapid system recovery from power transients with minimum loss and power interruption specified in MIL-E-16400, of traffic data other than that contained in messages being processed during power failures.

3.1.9 Illumination. - The illumination of all dials, controls, and lights shall be governed by good human engineering practice under conditions of equipment installation in an illuminated compartment aboard ship and shall be in accordance with MIL-E-16400.

3.1.10 Acoustic noise. - Acoustic noise shall be held to an absolute minimum and in no case be objectionable to the degree of hindering normal operator function within the immediate area. The airborne noise requirements of MIL-STD-740 shall apply.

3.1.11 Connection hardware. - All required mating cable connectors and cables within the communication center shall be furnished as part of the system. Stuffing tubes, if used, shall also be furnished in accordance with MIL-E-16400.

3.1.12 Codes, formats. - The system shall be capable of accomodating variable length messages transmitted or received in ITA No. 2 teletypewriter alphabet (American Version) (Baudot) or American Standard Code for Information Interchange (ASCII) which have been formatted in accordance with the procedures of Navy multichannel broadcast practice, DNC-5, ACP-127, and JANAP-128. Additionally, means for digital data mode operation may be provided upon definition thereof.

3.1.13 Average traffic units. - System capacity and processing rate requirements herein shall be based on an average message length of 200 words (1200 characters of up to eight bits per character).

3.1.14 Data rates and capacity. - The system when operating "on-line" shall accomodate standard teletype data rates and such other rates as specified in MIL-STD-188 up to 2400 baud. It shall be possible to handle up to 2500 average messages per day and to retrievably store up to 7500 messages.

3.1.15 Precedence. -

3.1.15.1 The MPDS shall recognize and process four levels of precedence as follows:

Designation	Prosign
FLASH	Z
IMMEDIATE	O
PRIORITY	P
ROUTINE	R

3.1.15.2 Distribution by precedence. - The processing time requirements in 3.11.15.1 are based on the following distribution:

FLASH	Less than 1 percent
Immediate	Less than 10 percent
Priority	Less than 30 percent
Routine	Not less than 60 percent

3.1.16 System processing rates. -

3.1.16.1 Automatic processing time requirements are mean times based upon the average message of 1200 characters. The mean processing time for all messages shall be less than 4 seconds. Not more than

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one in 10^3 FLASH messages shall have a processing time over 6 seconds. Not more than one in 10^3 messages of other precedences shall have a processing time over 8 seconds. These time limits shall be based on the distribution specified in 3.1.15.2.

3.1.16.2 The time to locate a message in the on-line retrieval store and display the message for inspection on the console display shall not exceed 2 seconds, as a design goal.

3.1.16.3 The required switching to bypass the central processor for direct distribution of messages to user terminals shall be capable of accomplishment in 1 minute or less.

3.1.17 Computer programs. - An operating program and necessary service programs shall be supplied with the system. Computer programs shall be modular in design such that addition, deletion, or change of function(s) can be made with a minimum of reprogramming. The operating program shall provide for the automatic accomplishment by or under control of the central processor, of system operations as specified in 3.2.2.

3.1.18 Fault detection, location, isolation, and recovery. -

3.1.18.1 It shall be a design objective of this system to provide the capability to detect fault conditions in the MPDS equipment as they occur, or before such faults can adversely affect programs or data.

3.1.18.2 Means shall be provided by diagnostic programs or otherwise, to locate faults in the MPDS equipment as soon as possible after a fault condition has been detected. It shall be a design objective of this system to locate faults to at least the system equipment module level, and preferably to the replaceable card level. Faults thus located shall be isolated if possible for off-line repair.

3.1.18.3 It shall be a design objective of this system to provide to the extent practicable within time and funding limitations, on-line fault recovery capability by means of which the system is reconfigured to isolate faulty equipment modules and retain good modules, while preserving the data base. The system shall be capable of utilizing such fault recovery programs as they become available.

3.1.19 Security protection. - Provisions shall be made to handle EFTO, Confidential, Secret, and Top Secret messages in the message center, in the communications channel terminations, and at designated user terminals without risk of compromise in accordance with Bureau of Ships Instruction 05510.33. The provisions of FED-STD-222 shall apply to the maximum extent practicable. (Copies of Bureau of Ships Instruction 05510.33 may be obtained from Commander, Naval Ship Engineering Center, Sec. 6178D.)

3.1.20 Equipment separation limits. - The equipment shall be capable of operation at remote distances as follows:

(a) Between the control console and the data processor and internal distribution switching up to 100 feet.
(b) Between the data processor and the bulk storage, high-speed printer and the internal distribution switching equipment up to 100 feet.
(c) Between the internal distribution switching equipment and user terminal equipment up to 1000 feet.

3.1.21 Dependence of facilities control system. - The MPDS shall be completely dependent on the Facilities Control System (FCS) for the provision of good communication channels. Forward error detection and correction shall be provided on up to 16 receive and eight transmit channels. Indication of communication channel status at the MPDS control console shall be kept to a minimum.

3.2 System. -

3.2.1 System description. - The major units of the on-line MPDS shall be as shown on figure 1. The intent is to utilize equipment units or designs which are in being and readily available to the greatest extent possible. A listing of major equipment units and their availability status is specified hereinafter.

3.2.1.1 The central processor shall be an existing general-purpose (stored program) computer designed for shipboard use.

3.2.1.2 The multiplexers shall be existing general purpose (stored program) computers designed for shipboard use, and shall divide between them the special multichannel logic assemblies for simultaneous

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connection of the dispersed user terminals to the processor. Additional multiplexer channels shall connect communication receiving/transmitting channels (data and control lines as required) to the processor for "on line" operation.

3.2.1.3 The bulk storage unit shall be an existing magnetic disk unit designed for shipboard use, closely associated with the computer and multiplexers. Messages, log entries, and accounting information shall be stored in it and printed out for permanent file.

3.2.1.4 An MPDS control console shall house high-speed tape reader/punch, CRT displays, keyboards, and printers, plus communication receiving/transmitting channel status and user terminal status indicators.

3.2.1.5 The user terminals shall be compact consoles housing keyboard, printer, and CRT display units and special controls and indicators, or, in certain of them, card reader/punch.

3.2.1.6 A high-speed line printer, designed for shipboard use shall be provided for rapid print out of messages, logs and indices, as well as for diagnostic use.

3.2.1.7 Provisions shall be made for connecting and switching printing reperforators, page printers, and paper tape readers for terminating communication channels as backup in the event of malfunction of the automatic system.

3.2.1.8 Interconnection facilities which can be controlled automatically, semi-automatically, and manually shall be provided.

3.2.1.9 Storage and retrieval capabilities for long-term files shall be provided.

3.2.1.10 The interface between the MPDS central processor and the FCS circuit sensing multiplexer, to provide for input of the communication circuit sensing signals to the MPDS central processor, will be a standard I/O channel of the central processor.

3.2.1.11 A continuous or periodic indication of suspected channel trouble shall be provided to the Facilities Control Console (FCC) and MPDS Control Console for those channels being processed automatically

3.2.2 System operation. -

3.2.2.1 The following operations (except backup provisions in 3.2.2.5.2) shall be performed when the MPDS is operating on-line with respect to the communication channels. During on-line operation, messages enter the system via multiplexer channels. In the process of message entry, such information as time of receipt, format indicator, and distribution indicator(s) shall have been added to each message either manually or automatically.

3.2.2.2 Incoming message processing. -

3.2.2.2.1 The operations detailed herein for processing incoming messages shall include broadcast message screening, duplicate search, indexing, logging, retrievable storage, and internal distribution.

3.2.2.2.2 Essentially the same operations shall be performed on all messages entering the MPDS, whether from communication receiving channels for local distribution or relaying or both, or from user terminals for transmission or local distribution. Certain operations, however, such as duplicate message search, will not be performed on locally originated messages.

3.2.2.2.3 Broadcast message screening. - All messages arriving on Fleet Broadcast channels shall be screened automatically and logged, but not otherwise processed, stored, or distributed, except for those on a preselected changeable list of addressees.

3.2.2.2.4 Duplicate message search. - A search index compiled for each message from precedence, date-time group, originator, and storage location shall be compared with indices of all other messages held in bulk storage to determine if it duplicates the date-time-group and originator of any of them. For relayed or readdressed messages, a second search index based on the station from which received in lieu of originator shall be compiled and similarly compared. Indices shall be retained until the messages are purged from bulk storage. If no duplication is found, the processing sequence shall continue automatically. If duplicate indices are detected the processor shall store them and cause a display or other readout. Communicator

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action will then be necessary to either drop the incoming message or continue its processing, with or without (optional) making a visual comparison of the messages to confirm the duplication.

3.2.2.2.5 Message logging. - Message logging information shall be compiled and compactly recorded in bulk storage. This information shall include the following:

(a) Broadcast serial
(b) Precedence
(c) Date-time group
(d) Originator
(e) Time of receipt
(f) Time of delivery
(g) Internal distribution data
(h) System serial number
(i) Classification
(j) Content indicator
(k) User terminal sequential number

It shall be possible, either periodically or on demand, to print out all or selected portions of the log information in compliance with doctrine or command procedure.

3.2.2.2.6 Message distribution. - Message distribution to user terminals shall be performed automatically based on the content indicator code prescribed by JANAP-128. Messages received in ACP-127 or other format (without content indicator) shall be displayed to the communicator for visual inspection and keyboard insertion of the JANAP-128 content indicator to enable distribution.

3.2.2.2.6.1 The central processor operating program shall provide for a local distribution table changeable by keyboard or paper tape. This table shall list, for each category in the content indicator code, the one or more user terminals to receive the message, and the number of copies to each terminal, as determined by the local command. In each case the "action" terminal, if action is required, shall be listed first or be otherwise indicated. The presence of the content indicator in the heading of the delivered message makes unnecessary, except in special cases, adding local distribution information in each message.

3.2.2.2.6.2 Positive safeguards shall be provided against unauthorized disclosure of classified information. A coded device in the personal possession of authorized personnel shall be employed at user terminals to establish for the MPDS control console operator the clearnace level of the terminal and, when necessary, the identity of the person.

3.2.2.2.6.3 Accountability will generally be accomplished by a sequence numbering system. No user terminal acknowledgement is required except for top secret messages. Communications center log will contain distribution and sequence numbers. Each individual user terminal is responsible for all copies of each message received.

3.2.2.3 Message retrieval. - The bulk storage unit shall provide capacity to store up to 3 days of traffic at peak rates. Any message in this store shall be retrievable from the MPDS control console or any user terminal provided they have the proper clearance and need to know. Accountability shall be the same as in 3.2.2.2.6.3.

3.2.2.3.1 Retrieval of messages in bulk storage shall be accomplished by keyboard entry of the message system serial number or date-time group and originator if known; then the designations of the terminal (s) to receive the message and number of copies to each terminal; and finally initiation of the retrieval action. The processor shall provide direct readout to the designated terminal(s).

3.2.2.3.2 It shall be possible to alter the above sequence by requiring the processor first to display any retrieved message at the MPDS control console, with manual action required to enable the requested terminal distribution, to guard against unauthorized disclosure.

3.2.2.3.3 Retrieval of messages more than 3 days old from long-term files shall be done manually.

3.2.2.4 Outgoing message processing. - Drafting of messages shall be performed at user terminals by means of the keyboard and CRT display. Message preparation shall affect only the terminal concerned until the completed message is entered in the central processor by manual control.

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3.2.2.4.1 Upon entry, the message shall be indexed, logged, checked for precedence and classification, and finally for release authorization indicator. If the latter is not present, the message shall be displayed at the user terminal designated for release authorization.

3.2.2.4.2 Should release be denied, comments as to the reasons or any changes desired can be added by keyboard at the authorization terminal and the message directed back to the originator. If release is authorized, the proper indicator is added. This shall enable the central processor to proceed with transmission preparation, such as additional indexing, logging, checking for precedence and classification, and insertion of date-time group.

3.2.2.4.3 The processor shall interpret each addressee entered by the originator and, by means of table look-up in readily changeable tables, add call signs, prosigns, and routing indicators from ACP-117 to complete the message heading, and select the proper transmitting channel(s). The message is then sent over that channel in first-in, first-out order by precedence. Should any operator action be required to complete these functions, the processor shall alert the MPDS control console operator, and provide an indication of the action(s) required.

3.2.2.4.4 Outgoing message processing shall be completed by distribution to the originator and any other local terminals indicated, and by completion of log entries to include these actions.

3.2.2.5 Message entry exit. -

3.2.2.5.1 Automatic message Entry/Exit operation. - The CVA(N)-68 MPDS shall be connected on line with the communications channels to be serviced, as shown on figure 1. Multiplexer capability shall be provided for twenty each receive and transmit active communication channels in addition to up to twenty user terminals. (For capacity and rates see 3.3.5.3).

3.2.2.5.1.1 The multiplexer channels shall provide serial-to-parallel and parallel-to-serial conversion, buffering, and all other functions necessary for on-line operation.

3.2.2.5.1.2 Simultaneously with transfer of traffic into the central processor, the traffic shall be transferred directly to the bulk storage unit for redundant storage to avoid loss in the event of processor malfunction. A bulk storage area shall be assigned for redundant storage which is equivalent to 5 minutes of traffic with all incoming lines busy. This storage shall be recirculated by dropping oldest traffic first.

3.2.2.5.1.3 It shall be possible and permissible to pre-enter manually into the processor certain significant traffic parameters, such as broadcast designators, heading format designators, data rate, and so forth, characteristic of each connected channel. This will simplify processing and make unnecessary including this information with each message.

3.2.2.5.1.4 During automatic on-line operation of MPDS it shall be possible to enter incoming messages manually, by means of tapes at the MPDS control console, into the central processor for automatic processing and distribution.

3.2.2.5.1.5 Means shall be provided for correcting messages at the MPDS console CRT/keyboard.

3.2.2.5.2 Backup provisions. -

3.2.2.5.2.1 Backup for normal MPDS operation shall occur in several steps determined by the nature of the failure. In no case should a single equipment failure completely disable automatic operations.

3.2.2.5.2.2 In the case of sufficient MPDS failures to cause the cessation of automatic operation, it shall be possible to switch up to twenty incoming communications channels to reperforators or page printers, or both visually screen the messages, and enter them manually into the MPDS tape readers, which may be used for internal distribution. Additional data regarding internal distribution shall be entered at the keyboard by manual switching. It shall be possible during such periods to switch up to twenty outgoing communication channels to tape readers for manual entry of outgoing message tapes. Logging, indexing, retrieval, storage, and automatic distribution by content indicator for incoming messages will not be possible during such periods. Messages processed by manual tape relay operation during such periods will be re-entered for logging, indexing, and retrieval storage after the resumption of automatic operation.

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3.2.2.5.2.3 It shall be possible to provide copies of messages for messenger distribution by means of a duplicating machine, and original copies from up to 20 backup page printers.

3.2.2.6 Service to facilities control console. -

3.2.2.6.1 The MPDS central processor shall store and process facilities control information input from and output to an FCC user terminal (CRT display, keyboard, and medium speed printer). The central processor shall contain a list of all communications equipments with their status and interconnections with other equipments within a communications channel. The equipment status and circuit configuration shall be entered at the FCC user terminal. The circuit configuration shall be checked by the central processor by sensing inputs from the FCS circuit sensing multiplexer. All active facilities control information shall be available to the FCC on request of the operator. The complete list of equipment status and circuit configuration shall be printed periodically at the FCC user terminal for backup in case of failure of the MPDS central processor.

3.2.2.6.2 Calculations of self-interference of medium and high frequencies (2 to 32 Mc.) to be used in the communications plan, supplied from the FCC, will be performed by the MPDS processor and displayed at the FCC on request of the operator.

3.2.2.6.3 A continuous or periodic indication of suspected channel trouble shall be provided to the FCC and MPDS Control Console for those channels being processed automatically.

3.3 System composition. - The CVA(N)-68 MPDS shall be comprised of the units specified hereinafter.

3.3.1 Central processor. - The central processor shall be a CP-855/UYK or similar member of the NTDS family of shipboard approved processors.

3.3.2 Bulk storage. - The bulk storage shall be a multi-drive disk storage unit, RD-281/UYK or equivalent bulk storage device with a capacity of at least 100 million bits.

3.3.3 MPDS control console. - The MPDS control console shall be designed to occupy a minimum of space consistent with good human factors engineering, taking into consideration the operations to be performed, functional grouping of equipment and controls, and maintenance. Controls, indicators and panel layout and selection of colors shall be in accordance with MIL-E-16400. The MPDS control console shall contain the elements specified hereinafter.

3.3.3.1 High speed paper tape reader. - The reader shall be capable of reading 5, 6, 7, and 8 level fully perforated paper tape at a rate of at least 240 characters per second.

3.3.3.2 High speed paper tape punch. - The punch shall completely perforate paper tape at a rate of at least 60 characters per second.

3.3.3.3 CRT/Keyboard. - The CRT/Keyboard (two each) shall be identical with the CRT user terminals specified in 3.3.6.1.

3.3.3.4 Printer. - The printers (two each) shall be of the same type as those of the user terminals, specified in 3.3.6.1.

3.3.3.5 MPDS controls and indicators. -

3.3.3.5.1 Control devices shall be provided for the following:

(a) Interconnection facilities (see 3.3.7).
(b) Processing equipment, for effecting parameter changes in computer programs.

3.3.3.5.2 Indicators shall be provided for the following equipment and functions:

(a) MPDS equipment configuration, status, and alarms, including user terminals.
(b) MPDS processing alarms and thresholds determined by effects of the system data on the operating programs.
(c) A good/bad indication for each transmit and each receive channel (see 3.2.1.11), and a "hold traffic" indication for each communication transmit channel.

3.3.3.6 Parallel interface. - A parallel interface which will allow the units of the MPDS control console to communicate directly with the central processor shall be provided.

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3.3.4 High speed line printer. - The high speed line printer shall be an RO-280/UYK or equivalent capable of printing at least 600 lines per minute.

3.3.5 Multiplexer. - The multiplexers (two required) shall be CP-848/UYK or similar members of the NTDS family of shipboard approved processors, and shall provide the functions of serial-to-parallel conversion, character and word assembly, and block buffering on incoming traffic and the inverse functions for outgoing traffic. Either multiplexer shall be capable of performing all required multiplexing functions.

3.3.5.1 Interfaces. - All serial interfaces shall conform to MIL-STD-188 low level interface requirements. A bit buffer shall be provided for each serial line. All parallel interfaces shall conform to the standard NTDS fast interface as defined in NAVSHIPS 0967-051-6040.

3.3.5.2 Rates. - The multiplexer shall be capable of receiving or distributing serial data at standard teletype data rates and other such rates as specified in MIL-STD-188 up to 2400 baud. Parallel data transfer (full or partial word) shall be capable of proceeding at a rate of at least 50,000 words per second. For serial internal distribution and collection, message data will be transferred to and from the user terminals synchronously by the multiplexer.

3.3.5.3 Capacity. - Provisions shall be included for terminating twenty active receive and twenty active transmit serial communication channels, and twenty half-duplex user terminal channels.

3.3.5.3.1 Communications channel termination capacity shall be limited to a mix of high speed (1200 or 2400 baud) and low speed standard teletype rate channels, the total rate not to exceed 16,800 baud.

3.3.5.3.2 Message distribution and collection channels shall be made up of a mix of high and low speed channels, the total rate not to exceed 43,350 baud.

3.3.6 User terminals. - The maximum of twenty user terminals shall be capable of both receiving and generating traffic. They may be of two types: (a) CRT/keyboard/printer and (b) card reader/punch. The interfaces shall conform to the MIL-STD-188 low level interface requirements.

3.3.6.1 CRT/Keyboard/printer terminal. - The CRT/Keyboard/printer terminal shall be designed modularly so as to allow flexible use in various combinations such as the printer either by itself for receipt of messages only or in conjunction with the keyboard for message preparation. The terminal shall be capable of operating synchronously at MIL-STD-188 rates up to 2400 baud. Designated user terminals shall be capable of printing both communication and weather upper case symbols. The terminal shall provide local message storage for at least 1440 characters.

3.3.6.1.1 The CRT unit shall be capable of displaying at least 20 lines of 72 characters per line.

3.3.6.1.2 The keyboard shall conform to MIL-STD-188 either ASCII or ITA#2 (American version) character set and layout.

3.3.6.1.3 The terminal shall be provided with a positive security device which must be activated before the terminal can be used. The clearance status of this device shall be capable of interrogation by the processor.

3.3.6.1.4 Indicators shall be provided for existence and precedence of messages awaiting delivery.

3.3.6.1.5 The CRT/keyboard portion of the FCC user terminal shall fit in a 19-inch cabinet in accordance with MIL-STD-189.

3.3.6.2 Card terminal. - This terminal, if required, shall contain a card reader and a card punch.

3.3.6.2.1 The card reader shall be capable of reading at least 400 80-column cards per minute. The punch shall be capable of punching at least twenty cards per minute.

3.3.6.2.2 The terminal shall contain storage adequate to hold data equivalent to two full cards.

3.3.6.2.3 Controls and indicators shall be provided similar to those of the CRT/keyboard/printer terminal.

3.3.6.2.4 Operating rates of standard teletype rate 1200 and 2400 baud shall be provided.

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3.3.7 Interconnection facilities. - The interconnection facilities shall consist of automatic, semiautomatic, and manual switching required to interconnect the user terminals, communication channels, multiplexers, reperferators, paper tape readers, keyboards, page printers, the bulk storage device, and the MPDS control console.

3.3.7.1 All interconnection facilities shall be treated as RED, and no BLACK data, control, or indicator signals, as defined in Bureau of Ships Instruction 05510.33, shall be included. All signals including control shall be MIL-STD-188 low level.

3.3.7.2 The following interconnection capabilities shall be provided:

(a) Any communication channel to either multiplexer (on-line operation).
(b) Any user terminal to either multiplexer (on-line operation).
(c) The bulk storage device to the central processor (on-line operation) or either multiplexer (backup operation).
(d) Any incoming communication channel to any reperforator and/or page printer (on-line operation or backup operation).
(e) Any outgoing communication channel to any standard teletype rate paper tape reader (backup operation).
(f) Any incoming communication channel to either of the two CRT/printer operating stations at the MPDS control console (backup operation).
(g) Any outgoing communication channel to either of the two CRT/keyboard operating stations at the MPDS control console (backup operation).
(h) Any standard teletype rate paper tape reader or either MPDS control console high speed paper tape reader to any number of user terminals (backup operation).

3.3.7.3 Switching. - Switching shall be as follows:

(a) Automatic switching control signals for the interconnection facilities shall be originated by the central processor; semiautomatic control signals shall be originated at the MPDS control console. Manual switching shall be performed at the interconnection facility.
(b) The degree of backup operation will depend on the equipment failure occuring, such as failures in the MPDS control console, the bulk storage, the multiplexer, the high speed printer, or the central processor. The central processor, if possible, should sense such failures and automatically initiate transfer from on-line operation to the appropriate degree of backup operation.

3.3.7.4 Indications of interconnections shall be provided to the MPDS control console and the FCC.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification: where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection. - Inspection of the equipment shall be performed in accordance with MIL-E-16400, except as otherwise specified herein.

4.3 Preproduction inspection. - Preproduction inspection shall be performed on a preproduction sample in accordance with MIL-E-16400, except that the accelerated life, water cooling, and salt spray test are not required.

4.3.1 Reliability. - Reliability test shall be conducted in accordance with MIL-R-22732, at test level 3, for 200 hours. The preproduction report should specify if there is any discrepancy between the reliability prediction and the reliability test.

4.3.2 The preproduction equipment shall be thoroughly reconditioned by the contractor before delivery as a production equipment.

4.4 Quality conformance inspection. -

4.4.1 Production inspection. - Production inspection shall be performed on each equipment in accordance with MIL-E-16400.

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4.4.2 Production control inspection. - Production control inspection shall be performed on samples selected as specified in 4.4.2.1 in accordance with MIL-E-16400, except that the reliability and water cooling tests are not required.

4.4.2.1 Sampling plan. - Where two or more equipment units (such as user terminals) are required, sampling for production control inspection shall be as follows: As units are produced and their number increases, samples shall be selected at random in accordance with table I. After a design change has been made, sampling shall be started with the first entry in this schedule and progress downward as for a new equipment design, unless otherwise approved by the Naval Ship Engineering Center.

Table I - Sampling for production control inspection.

Cumulative number of production units	Cumulative number of samples
2 to 4, inclusive	2
5 to 9, inclusive	3
10 to 15, inclusive	4
16 to 25, inclusive	5

4.4.3 Environmental inspection. - Environmental inspection shall be performed in accordance with MIL-E-16400 on one production user terminal equipment which has passed the production and production control inspection, except that the accelerated life and salt spray tests are not required.

4.5 Proposed tests. - A list of proposed tests, test procedures, and test equipment shall be submitted by the contractor to the Naval Ship Engineering Center for approval, 60 days prior to setting up test equipment for preproduction and production control inspection.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Preservation and packaging. - Preservation and packaging, which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules, and Regulations or other carrier regulations, as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation and packaging, packing, and marking for shipment shall be specified by the procuring activity:

(5.2.1 Preservation and packaging, packing, and marking. - The equipment, technical publications, accessories, and repair parts shall be preserved and packaged level A or C, packed level A, B, or C, and marked as specified (see 6.1) in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) Preservation and packaging, packing, and marking, requirements if other than specified in 5.1 (see 5.2).

6.2 Preproduction. - Invitations for bid should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Procuring activity:
Navy - SH
(Project 5895-N558Sh)

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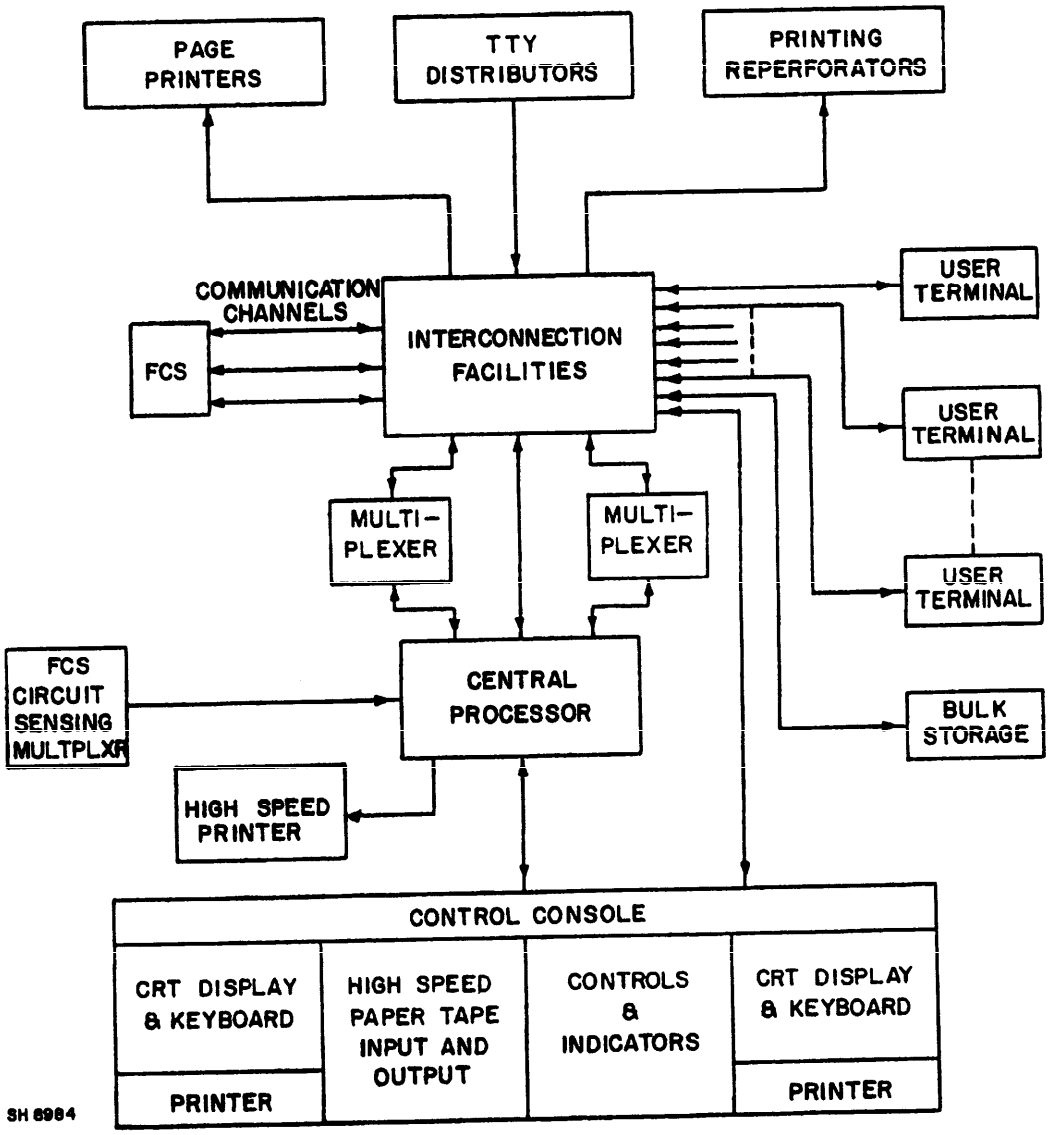


Figure 1 - Block diagram of Message Processing and Distribution System, for CVA(N) - 68.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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DEPARTMENT OF THE NAVY

Naval Ship Engineering Center
WASHINGTON, D. C. 20360

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